

Challenge: Mod 26

Category: Cryptography

Description: The challenge provided a text string which was encoded using ROT13, a specific type of Caesar cipher that shifts every letter by exactly 13 positions in the alphabet.

Approach: I used CyberChef's ROT13 module to decode the text. Using this module, the text was decoded in a single pass, since ROT13 shifts each letter by exactly 13 positions — revealing the flag immediately.

Tools used: CyberChef

Flag: picoCTF{next_time_I'll_try_2_rounds_of_rot13_45559abd}

What I learned: ROT13 is a special case of the Caesar cipher where the shift value (13) is exactly half the alphabet's length (26) — this means the same operation both encodes and decodes the text. It's historically used for lightweight obscuring (like hiding spoilers), not real security, since it's trivially reversible with any basic tool. The flag's own phrasing ('2 rounds of rot13') plays on this — applying ROT13 twice would undo the encoding entirely, since two 13-shifts complete a full loop of the alphabet.